

Product Requirements Document (PRD)

Smart College Canteen Food Ordering Platform

1. Project Overview

Project Name: Smart College Canteen Food Ordering Platform

Project Type: Web Application / Mobile Responsive Platform

Purpose: The platform digitizes and optimizes the food ordering process in college canteens by reducing queues, improving order management, enabling digital payments, and streamlining kitchen operations.

2. Problem Statement

Traditional college canteens face issues such as long queues, token confusion, delayed food preparation, manual billing errors, and inefficient order management. This platform aims to solve these problems using automation and real-time management.

3. Objectives

Primary Objectives:

- Reduce waiting time
- Digitize ordering
- Improve kitchen efficiency
- Enable secure payments

Secondary Objectives:

- Analyze food demand
- Reduce wastage
- Improve user experience

4. Target Users

Primary Users: Students, Faculty, Canteen Customers

Secondary Users: Admins, Kitchen Staff, Cashiers

5. User Roles

- Student – Order food, track orders, payment
- Admin – Full platform control
- Cashier – Billing and payment handling
- Kitchen Staff – Preparation management

6. Functional Requirements

6.1 Landing / Welcome Page

- College branding
- Today's specials

- Food categories
- Operating hours
- Login button
- Optional live crowd indicator

6.2 Student Login Page

- Login using roll number/register number
- Password or OTP authentication
- Fetch student details automatically
- Prevent fake users

6.3 Student Dashboard

- Current token status
- Recent orders
- Notifications
- Wallet balance (optional)

6.4 Food Menu Page

- Breakfast, Lunch, Snacks, Juice categories
- Veg/Non-veg filters
- Price display
- Real-time availability status

6.5 Food Item Detail Page

- Food image
- Description
- Price
- Quantity selector
- Optional add-ons and spice level

6.6 Cart Page

- Selected items
- Quantity modification
- Bill calculation
- GST calculation
- Estimated preparation time (optional)

6.7 Checkout Page

- Order confirmation
- Payment method selection
- Generate bill number
- Generate token number
- Generate QR code

6.8 Live Order Tracking Page

- Order received
- Preparing
- Ready for pickup
- Completed

6.9 QR Pickup Verification Page

- Student presents QR code
- Staff scans QR code
- Order marked delivered
- Prevents duplicate collection and fraud

6.10 Order History Page

- Previous orders
- Bills
- Reorder functionality
- Analytics support

7. Admin/Canteen Modules

7.1 Admin Login Page

- Secure login
- Role-based access control
- Admin, Cashier, Kitchen Staff roles

7.2 Admin Dashboard

- Total orders
- Revenue overview
- Pending orders
- Popular food items
- Charts and analytics

7.3 Order Management Page

- Accept/reject orders
- Mark preparing
- Mark ready
- Mark delivered
- Display token number and student details

7.4 Kitchen Dashboard

- Pending order queue
- Preparation timers
- Priority orders
- Real-time updates

7.5 Menu Management Page

- Add/edit food items
- Upload images
- Disable unavailable items
- Instant stock control

7.6 Student Management Page

- Student database
- Roll number management
- Blocked users
- Fine tracking (optional)

7.7 Billing & Reports Page

- Daily sales reports
- Revenue analytics
- Food sales analysis
- PDF invoice generation

8. Advanced Features

- Crowd prediction system
- Pre-order scheduling
- Smart notifications
- Feedback and complaint system

9. Non-Functional Requirements

- High performance
- Scalability
- Secure authentication
- Reliable order handling
- User-friendly UI

10. Technology Stack

- Frontend: HTML, CSS, JavaScript / React
- Backend: Node.js / PHP
- Database: MySQL
- QR System: QR Generator API
- Notifications: Firebase / Telegram API

11. Database Requirements

Main tables include:

- Students
- Orders
- Payments
- Food Items
- Notifications
- Feedback

12. Workflow

Student Workflow:

Login → Browse Menu → Add to Cart → Payment → Receive QR & Token → Pickup

Admin Workflow:

Receive Order → Prepare → Mark Ready → Scan QR → Deliver

13. Security Requirements

- Encrypted passwords
- Secure authentication
- QR validation
- Role-based access control

14. Expected Outcomes

- Reduced queue congestion
- Improved canteen efficiency
- Better student experience
- Reduced manual errors

15. Future Enhancements

- AI-based recommendations
- Facial recognition login
- Mobile application
- Smart inventory prediction

16. Conclusion

The Smart College Canteen Food Ordering Platform modernizes traditional canteen operations using digital ordering, QR verification, and centralized management to improve efficiency and user experience.